J. DRUG EDUCATION, Vol. 17(1), 1987

SMOKING PREVENTION PROGRAM FOR CHILDREN: A REVIEW

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ABSTRACT

Since smoking is recognized as one of the most significant health hazards, researchers have directed a lot of energy in combating this health hazard, in particular in the prevention of children taking up smoking. Health education has been advocated as the most effective preventive approach to this problem, however, the findings from this approach are equivocal. This article 1) reviews the literature regarding factors associated with children's initiation into smoking, and 2) examines the efficacy of health education programs in preventing smoking in children. It is concluded that while health education programs using peer leaders as health educators have been reasonably successful at reducing smoking rate and onset, parent-implemented health prevention programs aiming at children at younger age, may be more effective in reducing the rate and onset of smoking in children.

Smoking is a public health problem of enormous proportions which costs this, and other countries, thousands of lives and millions of dollars [1]. Over the past twenty years, there has been a steady accumulation of data about the effects of cigarette smoking. Between 1964 and 1979, 24,000 major articles on smoking and health appeared in the scientific literature [2]. It is difficult to think of any other major medical subject on which opinions are so united. Today, there is no argument—smoking is clearly a health hazard and a killer. According to Doll, the avoidance of smoking would alone reduce the mortality from all cancers by about one-third (including avoidance of not only the large majority of cancers of

Preparation of this article was partially supported by a Special Project grant to Dr. T.P.S. Oei.

doi: 10.2190/V5Q3-G0TK-XMWX-G0Q0
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the mouth, throat and lung, but also a substantial proportion of the deaths attributed to cancers of the bladder, kidney and pancreas) [3]. It would almost eliminate chronic obstructive lung disease and the complications of peripheral vascular disease. It would reduce the age-specific mortality from aortic aneurysm by at least three quarters, and the mortality from myocardial infarction by about one quarter. At present, cigarette smoking is the single-most important preventable behavioral factor contributing to illness, disability, and death in the Western world [4].

Since the first Surgeon General's report on Smoking and Health in 1964, various measures have been taken in the interest of reducing cigarette smoking. Primary efforts include the banning of television advertisements, the development of smoking cessation programs, mandatory warnings to all cigarette smokers, and the initiation of smoking education programs in public schools. The proportion of cigarette smokers among the adult population has shown some decrease since the institution of these efforts, yet no such decrease has been evidenced in the proportion of children who smoke. The prevalence of cigarette smoking among teenagers remains high [5], and the relationship between smoking and respiratory symptoms is beginning to become obvious even at the primary school level [6, 7]. This finding underlines the importance of smoking prevention among this age group. Such prevention procedures must, however, be developed from a thorough understanding of the initiation process.

The first of the major goals of the present review was to carry out a detailed examination of children's initiation into smoking, looking specifically at their early experiences with and attitudes towards cigarettes. In considering this "initiation process" an investigation of the contribution of the most significant influential factors during this stage, namely that of parents and peers, was also carried out. To date, almost all attempts to scientifically investigate patterns of use among school children have concentrated either on those who have reached the high school level or, in the very rare case, those in the younger age bracket of ten to twelve years (for example: [8-10]). Indications from these studies and others are that for many children, initiation into the use of tobacco occurs at an even younger age [11, 12].

The second major goal was to review the literature on the efficacy of education programs aimed at changing the onset rate of smoking and attitudes towards cigarette use among these children.

In a smoking longitudinal study carried out on a sample of 787 nine-year-old Dunedin school children, findings indicated that parents had the greatest single influence on the attitudes and behaviors of their offspring in relation to smoking at this age, and yet discussed issues related to smoking infrequently with these children [11, 12]. The present article also reviews the literature regarding the roles of parents as health educators in the education programs (for example [4, 13, 14]).

PREVALENCE OF CHILDREN'S SMOKING

It appears from previous studies reporting on the prevalence rates of children's smoking that a very sizeable fraction of today's teenage and preteenage populations is electing to expose itself to the enhanced risk of cardiovascular disease, to greater likelihood of contracting cancer, and to other sources of death and injury arising from the use of cigarettes. While many of these studies have been concerned only with distinguishing between the smokers and the non-smokers or have concentrated on such data as weekly intake, a number of others have also provided information on the children who have experimented with cigarettes. One of the striking observations when we consider all findings, is that there is a steady rise with age in the proportion of children becoming involved with smoking. Indeed, Davis and Stavey have suggested that by the age of fourteen or fifteen, smoking or non-smoking behavior is an established pattern, and little experimentation takes place thereafter (cited in [15]).

Some of the youngest school children to participate in a smoking survey reported to date, were those involved in the study carried out by Egan [16]. Of the 787 nine-year-old Dunedin children who responded to questionnaires, around a third had tried puffing on a cigarette. Puffing was, however, less popular among Canadian school children in the grades four to six (nine to twelve years) sampled by Pederson, Stennet and Lefcoe, of whom only 25 percent had tried puffing on a cigarette [17]. Consistent with these findings, were those of Aitken who found that one-quarter of the 382 ten-year-olds studied from Central Scotland had also experimented [15]. One-third of the 76,000 ten to eleven and one-half-year-old Derbyshire students sampled by Bewley, Halil, and Snaith had tried smoking [6], as had one-third of the twelve-year-olds studied by Aitken [15]. Trying out smoking was most popular among almost 5,000 ten- to twelve and one-half-year-old Kent children [7] and around 6,000 ten- and eleven-year-old New South Wales school children—almost half of whom had had a puff [18].

A number of studies have drawn on children within the twelve- to thirteen-year age range to use in their samples. Altogether, just over half of the seventh-grade South Dakota and Iowa subjects in Palmer's sample of almost 3,000 children, had tried smoking [19]. Just under half of the sixth- to eighth-grade students sampled by Chassin, Corty, Presson, Olshavsky, Bensenberg and Sherman had had a puff [20]. Similarly, 50 percent of seventh-grade children from Minnesota sampled by Hurd, and colleagues [21], 41 percent of seventh-grade Minnesota children sampled by McCaul and colleagues [22], and 43 percent of Arkin et al.'s seventh-grade subjects from eight United States Twin Cities [5], could all be classified as puffers. Over 60 percent of the Australian twelve-year-olds studied by O'Connell et al. had at least some experience with cigarettes [18].

Although only 42 percent of Aitken's Scottish fourteen-year-olds admitted to having tried smoking [15], it appears that other high school children have experimented a lot more frequently. Among the American children thirteen years and older sampled by Palmer and Chassin et al., only around one-third of the children had not at least puffed on a cigarette [19, 20]. In a comparable study carried out in New Zealand on around 200 fourteen- to sixteen-year-olds, this "non-puffer" figure was reduced to 19 percent [10].

While the distinction between those who have, and those who have not experimented with cigarettes is clear in all the studies mentioned so far, it is here that the consistency with regard to definitions ceases. Despite this, however, it is apparent that few of those children in the twelve-years and under age range, have taken on smoking as a regular habit. Only 2.6 percent of the Dunedin nine-year-olds reported smoking in the last month [11,16], and none of the ten-year-olds and 2 percent of the twelve-year-olds in Aitken's [15] study of Scottish school children described themselves as "current smokers." Very few had even smoked more than one cigarette. The New South Wales school children in O'Connel et al.'s study puffed more frequently, with around 3 percent of ten-year-olds, 5 percent of eleven-year-olds and 9 percent of twelve-year-olds having smoked in the past weeks [18]. Similarly almost 6 percent of the eleven-year-olds in Leeder, Peat, Woolcock, and Blackburn's study of Sydney school children, were smoking on a weekly basis [23].

Among the Derbyshire and Kent children in the studies carried out by Bewley and colleagues, very few (almost 5%) were smoking on a weekly basis [6, 7]. Of the nine- to twelve-year-olds sampled by Pederson, Baskerville and Lefcoe 1.3 percent were smoking almost every day, and a further 4.3 percent smoked occasionally [24].

As with experimentation, more regular use of cigarettes increases steadily with age. Around 6 percent of Arkin et al.'s United States seventh graders (12 to 13 years) [5] and 8 percent of Leeder et al.'s Sydney twelve-year-olds, smoked weekly [23]. Similarly, 8 percent of Chassin et al.'s sixth- to eighth-grade Midwestern children were puffing more than once a month [20], and 8 percent of Hurd et al.'s seventh graders smoked twice a month or more [21]. Of the seventh graders in McCaul et al.'s sample, around 3 percent smoked at least weekly and a further 3 percent "occasionally" [22].

At the high school level, "regular" smoking becomes much more prevalent. In the younger (12 to 15 year) age group studied by Palmer, over 5 percent claimed to be smoking "most of the time" [19]. Similarly, 6.5 percent of Rabinowitz and Zimmerli's New York seventh- to ninth-grade children were smoking between five and ten cigarettes a day, with a further one-fifth who smoked less than five a day [25]. Just over 17 percent of the Dunedin high school students studied by Nye et al. smoked more than once a week, and another 20 percent of these children were "infrequent" smokers [10]. Seventeen percent of Wellington high school students studied by Beaglehole, Eyles, and Harding reported smoking less than two cigarettes daily and a further 15 percent described themselves as "heavy smokers" [9]. Over a quarter of Leeder et al.'s thirteen- to sixteen-year-olds were smoking on a weekly basis [23] and at least two-thirds of the Norwin sixteen- to eighteen-year-old high school students studied by Rudolph and Borland had smoked ten cigarettes or more in the past [26].

While absolute levels of smoking have been found to be similar between sexes in most studies, especially among intermediate and high-school age children [14, 22, 26], several researchers have noted the tendency for boys to begin experimenting with cigarettes at an earlier age than girls. In the study of 191 Dunedin high school pupils conducted by Nye et al., the boys surveyed were significantly younger when they first experimented with cigarettes (mean = ten years two months), than the girls (mean = eleven years seven months) [10]. Similarly, sex differences were reported by O'Connell et al. who found that only 38 percent of their males had never tried smoking, compared to 56.3 percent of the females [18].

In support of these findings are those of Zoller and Maymon who surveyed 1420 students from Israeli high schools, aged fourteen to eighteen years [27]. Results showed that the boys had started to smoke earlier than the girls. Similarly, of the 297 Brusselton school children aged twelve to fourteen years sampled by Colquhoun and Cullen, 11 percent of the twelve-year-old boys were classified as "smokers" compared to 6 percent of the girls [28]. However, by age fourteen, the proportion of girls and boys having tried smoking were very similar.

In summary then, it appears that while a regular pattern of cigarette use does not seem to become established until the teenage years, children begin experimenting with cigarettes at a much younger age. Between 25 and 50 percent of the primary school age children sampled in the studies reported, had at least puffed on a cigarette, with more boys experimenting in the younger age groups than girls. There was then a steady rise with age in the proportion of children trying out smoking. The large majority of high school students reported having had some experience with cigarettes, with around one-fifth smoking on a "regular" basis, and no significant differences across sexes.

ATTITUDES TOWARDS, AND REASONS FOR SMOKING

A consistency between children's attitudes towards smoking, and their actual smoking behavior has been noted by a large number of researchers. Attitudinal data were gathered from a sample of 955 high school students from the South Island of New Zealand in a study reported by Newman, Martin and Irwin [29]. They made use of a five-point attitudinal scale for each of 18 items. Results indicated that there was a significant difference between smokers and non-smokers on most attitude items. Smokers tended to agree that cigarette smoking is relaxing, pleasurable, and something nice to do when enjoying yourself. Non-smokers agreed that people would not smoke if they stopped to think about it, that parents should set a good example by not smoking, and that they would not let their children smoke if they were parents.

The major purpose of a study carried out by Downey and O'Rourke was to assess whether the initial attitudes and beliefs or a behaviorally homogeneous group could be utilized as indicators of future smoking behavior [30]. The students included were those seventh-grade Illinois youths (twelve to thirteen years) who classified themselves as never-smokers, and who participated in all three surveys over a two-year period. A total of 2096 met these criteria. Results showed that those individuals who maintained their non-smoking behavior during the two-year interval, initially were more oriented toward a non-smoking position, than those persons who initiated the smoking habit during the interim period.

The cigarette smoking habits and attitudes and associated social factors of 997 third- and fourth-form Wellington pupils (thirteen and fourteen years), were studied by Beaglehole, Eyles, and Harding [9]. The pupils were asked to indicate their response on a five-point scale, ranging from strongly agree to strongly disagree, to a number of statements on attitudes towards smoking. Results showed that over 80 percent thought smoking was a "waste of money," that "smoking causes cancer," and that "smoking is harmful to health." The majority of non-smokers and light smokers thought that one of the main reasons college students smoke is to be "part of the group," and that college students smoke to "show off," whereas the majority of heavy smokers agreed that "cigarettes are enjoyable" and that "smoking is something nice to do when you're having fun."

From the 7115 Derbyshire school children aged from ten- to eleven and one-half years described in the report by Bewley, Halil, and Snaith, a sample of 100 children previously identified as smokers (consume at least one cigarette a week), were taken [6]. All subjects were boys. These 100 were matched for sex, school class, and age with 100 experimental smokers (those who had puffed on a cigarette), and 100 non-smokers. Bewley, Bland, and Harris reported on these children's accounts of why others of their own age smoked or did not smoke, and their attitudes towards smoking [31]. Again, responses varied as a function of smoking behavior. More smokers than non-smokers said children smoke because they like it. Many more smokers (68%) thought children smoked to show off. About one-third of the children thought having friends who smoke was a reason, and that kids smoked to feel grown up. Both smokers and non-smokers said children of their own age did not smoke because of health worries and parents' disapproval. There was a significant difference between smokers and non-smokers in their descriptions of smoking as a dirty habit. Most children agreed that it was a bad idea to smoke, though less smokers (69%) thought so than non-smokers (88%). One-third of heavy smokers thought smoking was enjoyable compared to 9 percent of non-smokers. Just over half of the smokers and 92 percent of the non-smokers considered smoking to be a waste of money. While only 1 percent of non-smokers felt it was nice to smoke with friends, 45 percent of the smokers thought so. In summary, Bewley et al. claimed that the typical attitude of the smokers to smoking seemed confused, but generally negative [31]. The non-smokers on the other hand, seemed more definite in their attitudes to smoking.

Knowledge of the effects of smoking on health was measured as were attitudes towards smoking using semantic differential scales in O'Connell et al.'s study of 6224 New South Wales primary school children [18]. There were significant differences across smoking categories, with non-smokers having less favoring attitudes than children who claimed to smoke. Differences in knowledge between the categories of smoking behavior were also significant with regular smokers having lower scores.

Explorations of Palmer's 2729 South Dakota and Iowa junior high school students' feelings about smoking, were in the form of questions designed to assess reasons either for not beginning or for not continuing to smoke [19]. Major concerns included the effect smoking might have on athletic ability (especially for boys), concerns over how their parents might feel about it, and consideration of the potential serious disabilities, such as cancer. Similarly, the nine-year-old Dunedin school children interviewed in 1985 also most frequently associated smoking with harmful health effects and usually had negative feelings about cigarette usage in general [16].

In support of these findings, Levitt reported that the most popular reasons for not smoking among his sample of 51,507 fifth- to twelfth-grade Indianapolis school children were [32]: health reasons, not pleasurable, too young, too expensive, parental influence, undesirable effect, aesthetic reasons, it is a senseless behavior, no desire, negative effect on performance and undesirable reaction from smoking. Most popular reasons for smoking included pleasure, habit, peer influence, emotional involvement, to impress others, to imitate adults and to imitate siblings.

Newman et al. believed that the most significant result of their research was the indication that attitudes towards smoking did not appear to change with the age of the respondent [29]. The attitudes of thirteen- and fourteen-year-olds, both smokers and non-smokers, towards smoking were essentially the same as the eighteen-year-old students. They concluded that it appeared that attitudes towards smoking had been established prior to their high school years, and that the high school experience had little or no influence on attitudes towards cigarette smoking. Based on this observation, Newman et al. recommended that educators and health workers reassess the role of education in teaching about such potentially harmful habits as cigarette smoking. They saw a need to carefully consider educational programs designed to modify attitudes and behavior prior to the time of establishment.

Besides a significant association between children's attitudes towards smoking and actual smoking, a number of researchers have also noted a relationship between current smoking and future intention to smoke. Among Bewley et al.'s

Derbyshire children, 45 percent of heavy, and 40 percent of light smokers said they would be smoking in the future compared with 3 percent non-smokers [6]. Similarly O'Connell et al. reported that most of the children who claimed not to smoke did not intend to smoke in the future, while a far higher proportion of those who claimed to smoke intended to continue smoking [18]. Finally, in a report by the National Institute of Education adolescents' intentions to smoke cigarettes have been reported as the best prediction of their actual smoking status five years later Chassin et al. claimed that health beliefs are consistently correlated with behavioral intentions as well [20].

In summary, it seems that children's attitudes towards smoking are generally quite negative with most children agreeing with such statements as "smoking is harmful to health, a waste of money," and so on. Reasons given for smoking most commonly revolved around impressing peers, trying to act mature, and joining in as part of a group. Health worries, parental disapproval, lack of pleasure or desire, and negative effect on performance were popular reasons given for non-smoking. It appeared that most attitudes towards smoking were formed prior to high school and that there was a consistency between children's attitudes towards smoking and their actual smoking behavior. Those children who had previously experimented with cigarettes had more favorable attitudes towards smoking. In addition positive smoking attitudes and previous experience, were also associated with future intention to smoke.

IMPORTANT INFLUENTIAL FACTORS IN A CHILD'S DECISION TO SMOKE

Parental Influence

Parents play an important role in the growth and development of their children. They motivate and shape their children's attitudes and practices by the expression of their own attitudes and beliefs, as well as by their patterns of behavior. Such influence cannot be ignored or discounted during a child's formative years when decision making about specific behaviors becomes vital to normal development initially what a child comes to think about and do with cigarettes reflects in large measure, what the significant models in his family say and do with these drugs. They are the people whom he is likely to be exposed to first, longest, and most intimately, and if smoking is integrated into their lifestyle then it may quite logically follow that it should be integrated into his. Almost all studies based on this idea have consistently shown that family members play an important part in determining the smoking-related habits and attitudes of children.

Even among the early research investigating those factors influencing children's smoking over twenty years ago, the association of children's use of cigarettes with parental pattern of use was seen as "striking" [33]. These researchers administered questionnaires to 2823 seventh- to twelfth-grade Massachusetts high school children. In families where neither parent smoked, about one-quarter of the students were smokers; whereas in families where both parents were regular smokers approximately half of the children smoked. In families where only one parent was a smoker the proportion of smokers among students was nearly as high as when both parents smoked. The most striking differences were seen in the proportions of heavy smokers (more than five packs a week). In boys this proportion was more than twice as high, and in girls more than seven times as high in families in which both parents smoked, as compared with families where both parents were non-smokers.

Lemin claimed that it is usually accepted that parental example is a potent influence in determining whether children decide to smoke; and this was borne out in his study [34]. In all, 49.2 percent of the 918 fourteen-year-old Aberdeen (Scotland) school children he studied were classified as smokers (i.e., consumed cigarettes daily) if both of their parents smoked, compared to 8.3 percent if neither parent was a smoker.

The major purpose of an investigation by Nolte, Smith, and O'Rourke was to assess the relative importance of parental behavior and perceived parental attitudes about smoking, as factors possibly associated with whether youths chose to smoke [8]. The sample population consisted of 5409 grade seven to twelve students aged eleven to twenty years. A comparison between youth and parental behavior indicated that the youths' behavior was, in fact, associated with that of their parents. One-third of the smokers had two smoking parents. Not surprisingly, 53.1 percent of the never-smokers in contrast with only 26.6 percent of the smokers, reported that neither parent smoked. Students generally reported strong perceived parental opposition to their smoking. However, parents whose attitudes against youth smoking were stronger, were more likely to have children who smoked less.

Palmer's study was designed to assess the subject's perceptions of the conditions and forces present when he smoked for the first time. A total of 2729 junior high school students (aged approximately twelve to fifteen years), residing in South Dakota and western Iowa, were surveyed [19]. Results showed that parental smoking significantly affected that of their children, non-smoking children being much more likely to have two non-smoking parents than those who had tried smoking.

In a study conducted by Coe et al. in two public middle schools in St. Louis (grades seven and eight), an association was found between the smoking habits of adolescents and the smoking habits of their parents and friends [14]. Similarly, among the Wellington high school students studied by Beaglehole et al., the children were more likely to smoke if one of their parents smoked [35], as were the 1789 Norwin high school students sampled by Borland and Rudolph [36]. In support of these findings were those obtained by Revill and Drury, who surveyed all fourth-year secondary students in four New York schools (aged fourteen to fifteen years) [37]. They reported a significant relationship between the parents' and the children's smoking behavior.

When we turn our attention to those children in the younger primary school age group, this generational continuity in the use of cigarettes is just as strong, even though smoking at this age level is more frequently still very much at the experimental stage. Both the children and their parents were asked about the parents' smoking habits in the large Hunter Region study carried out by O'Connell et al. [18]. From this sample of 6224 ten- to twelve-year-olds, results of a child-parent analysis revealed that, as expected, children were much more likely to smoke if their parents did.

Bewley, Bland, and Harris also investigated some of the social factors which they felt might predispose children to start smoking at an early age [31]. This included both parents and peers. They found a highly significant association between the smoking habits of the 300 boys they studied, and those of their parents. Of the non-smokers, 40 percent had non-smoking parents, compared to 17 percent of experimenters, 19 percent of the light smokers (less than one cigarette a day), and none of the heavy smokers.

Finally, among the 787 Dunedin nine-year-olds there was a strong relationship between the parents' and their children's use of cigarettes [12]. Those children from homes where either mother or father used cigarettes were much more likely to have puffed on a cigarette themselves, than those who came from non-smoking households.

Peer Influence

Besides the important influence parental models have on a child's decision of whether or not to take up smoking, the pressure received from peer groups is another factor which is also critical in the onset of smoking [38]. That these two variables are of primary importance in adolescent smoking, was confirmed in a study of 2156 school children aged twelve to eighteen years carried out by Lauer, Akers, Massey, and Clarke, in Muscatine, Iowa [39]. Of those respondents who reported both parents as non-smokers, and whose best friends were also non-smokers, 80 percent had never smoked, and 3 percent were regular smokers. Of those with both parents and best friends who smoked, only 11 percent had never smoked, and 24 percent were frequent smokers. In addition, smoking was also related to anticipated or actual sanctions of parents and friends. Over half of those who expected or received a permissive reaction from parents, were occasional or regular smokers, and 68 percent of those who anticipated a discouraging parental reaction were non-smokers. Of those whose friends were seen as permissive, 53 percent were occasional or regular smokers, whereas 77 percent of those who anticipated a discouraging reaction from friends had never smoked. So it seemed that the influence of parents and peers was not only indirect through providing smoking or non-smoking models, but was also direct through their perceived rewarding or punitive reactions to the adolescents' smoking or non-smoking behavior.

The drive for conformity is influenced by the smoking habits of friends according to Lemin [34]. Among his fourteen-year-old Scottish high school students, over half of the smokers said that most of their friends smoked, while just under 10 percent of non-smokers had a majority of smoking friends. Only 2.5 percent of smokers had no friends who smoked, as compared with 36 percent of non-smokers. In addition, almost three-quarters of the smokers had obtained their first cigarette from a friend.

Included in the questionnaires given to the students in the Norwin high school study was an item designed to investigate the smoking habits of best friends [26]. From the data presented, it appeared that relatively few of the best friends of smoking students were not smokers themselves. Similarly, a great majority of the best friends of non-smokers were also non-smokers. Consistent with these findings were those of Palmer [19]. While half of the experimental smokers in his high school sample indicated that none of their friends smoked, only 13 percent of the boys who smoked "all the time" reported that none of their friends smoked. Less than a third of the non-smokers had smoking friends. The relationships were similar for girls.

An investigation was conducted by McCaul et al. in an attempt to determine whether a small set of variables drawn from previous research would predict adolescent smoking behavior one year later [22]. Subjects were 297 seventh graders enrolled in the Moorhead, Minnesota public school system. Strong support was obtained for the claim that peers serve as models who encourage or discourage smoking. The number of friends who smoked was a consistent, powerful predictor of future smoking behavior. McCaul et al. concluded that their findings indicated the potential importance of prevention activities to enhance adolescents' abilities to cope with peer pressure.

The influence of peer pressure on a child's decision to smoke is also apparent among those in the younger age group of nine to twelve years. In their analysis of the factors associated with smoking in their ten- to eleven and one-half-year-olds, Bewley, Bland, and Harris found that friendships both at school and outside, were significantly related to smoking [31]. Around three-quarters of the heavy smokers reported having school friends who smoked, compared with 37 percent of non-smokers. However, even more significant than these figures were those obtained in the Newcastle study of ten- to twelve-year-olds carried out by O'Connell et al. [18]. They concluded that friends' smoking behavior was the best prediction of the child's smoking status. Similarly, in their study of nine-year-olds, Oei, Egan, and Silva also investigated the relationship between peers' smoking habits and the children's own puffing experience [12]. Their results indicated that those children with smoking friends were much more likely to have tried puffing on a cigarette, than those whose friends were all non-smokers.

It seems from this review that those factors leading to the initiation and maintenance of smoking have been widely studied. Several main "causes" have been advanced, the two most important being familial modeling and peer pressure. While the exact mechanism for the parent-child smoking relationship is uncertain, and probably varied from case to case, it most likely includes identification with parents as role models, greater perceived or actual permissiveness regarding smoking, and increased opportunity for surreptitious smoking [40]. The pretext of peer pressure to smoke, on the other hand, is most likely related to the fact that smoking functions as a sign of independence from adult authority and of maturity. A youngster who refuses an offer of a cigarette may be called a "chicken" and may even risk being excluded from the peer group. The young person who accepts the offer gains not only greater acceptance from smoking peers, but also the appearance of being more "tough," "cool," or mature and adventurous, than his non-smoking peers [41].

Green and Green (cited in [42]) concluded that, in the vast majority of cases, peer relationships are the most powerful determinants of whether or not an adolescent will begin to smoke. For the pre-adolescent, however, other factors may be more salient in initiating experimentation with smoking. McCormick (cited in [42]) for example, reported that "... children's parents have a far greater influence on behavior than any other single factor, including their peer group." These claims possibly reflect a change of allegiance of the child from parent to peer groups as the child grows. Early adopters of smoking may be influenced originally by their observations of parents, and are later likely to be reinforced by peers' admiration of smoking as a "daring" activity. Unfortunately, as McAlister et al. have noted, given the dependence-producing nature of tobacco, there is a strong likelihood that its use will often continue after it has lost its social attractiveness with the peer group [40].

PREVENTION AS THE BEST POLICY—SMOKING EDUCATION PROGRAMS

It is unlikely that anyone would argue against the statement that, since it is difficult to break an established habit, it would be better, where the habit is considered detrimental to health, that it never be started (Newman cited in [43]).

Smoking cessation by adult smokers has been widely confirmed to be difficult and usually of short duration before relapse. In general, where assessments have been done, the results have not been particularly encouraging. Most studies report only 20 to 30 percent abstinence at three to six month follow-ups regardless of the techniques used [44]. Attempts at cessation of the smoking habit are accompanied by severe craving and withdrawal symptoms. Cessation is made difficult both by the element of physical and psychological addiction to nicotine, as well as the need to learn a new repertoire of non-cigarette related responses to daily events and stressors.

The recognition of the health consequences of long-term cigarette smoking and the difficulty of quitting once the smoking habit has been acquired, has led to increased attention being focused on smoking prevention programs for youth. Health education directed at children before the onset of addiction has been advocated as the most potentially effective method of preventing smoking-related disease. Because of this, there have been a number of attempts by schools, voluntary organizations and researchers, to prevent individuals from ever becoming cigarette smokers. With few exceptions, up until the late 1970s, the programs had attempted to dissuade children from smoking by providing them with factual information concerning the deleterious effects of cigarette use, or by using low-key, non-directive approaches, where the subject of smoking was raised "as appropriate."

In general such programs have shown little success. Rabinowitz and Zimmerli for example, reported on the effects of a smoking education program in which health hazards were chosen as the project's focus [25]. The program was administered to 785 grade seven to nine students. Results indicated that while there was evidence of knowledge gains and attitude change among the experimental group, there was no significant change in the subjects' reported use of cigarettes. Similarly, the smoking education program tested in the study by Beaglehole, Brough, Harding, and Eyles had no effect in reducing the use of cigarettes, or in changing the attitudes of pupils in the experimental school [35]. The study sample consisted of 997 Wellington high school pupils aged twelve to fifteen years, and the intervention program was one of the "non-directive approaches" where the subject of smoking was occasionally discussed with pupils. Educational resources included posters, pamphlets, tape recordings, charts, color slides, resource sheets, films, experiments, cartoons, values clarification exercises, and videotapes. An eight-month evaluation indicated the control and experimental groups did not differ significantly.

A complete review of such studies has been carried out by Eva Thompson who surveyed all reports of anti-smoking health education programs published between 1960 and 1976 [45]. She found that school-wide anti-smoking campaigns, youth-to-youth programs, and research on the effectiveness of different teaching methods, at best produced significant improvement only in the students' attitudes towards smoking. These programs did not produce improvement in behavior.

More recent intervention efforts have tended to be more successful than these traditional anti-smoking programs. The basic format for the more modern health educational efforts has still included a factual information component to establish an information base, which then sets the stage for the intervention effort. Most researchers seem to have accepted that this foundation of basic knowledge is necessary for the subsequent intervention effort to be effectively implemented [46]. This is reinforced by the findings of Bland, Bewley, Banks and Pollard who carried out an investigation into children's knowledge about some of the health consequences of smoking [41]. Data from their sample of 595 Derbyshire school children aged between eleven and thirteen years, suggested that many of the children who were experimenting with cigarettes did not understand what was meant by warnings against lung cancer. This was also true of other major diseases related to smoking.

However, while health-related education clearly is necessary to prepare the young person with reasons for not smoking, it is now recognized that the social pressures to smoke may actually override an awareness of the hazards of cigarette smoking and anti-smoking attitudes and beliefs. A sophisticated counter to the strong social influence encountered during adolescence is certainly needed. While almost all elementary school students will state that they do not want to start smoking, they may not be equipped to resist social pressure towards cigarette use, and thus may not be able to fulfill that intention as they enter the teen years [40]. More recently smoking prevention programs have focused on the social factors which appear to play a major role in influencing children to smoke. The objective of these programs has been to familiarize students with the nature of one or more of the major social pressures to smoke, and then to focus on building appropriate response repertoires in students, so that they learn to offset or neutralize these pressures to start smoking. These "innoculation" interventions have been successfully applied in several settings.

In spite of their conceptual similarities, however, these recent efforts have used quite different implementation procedures. Almost all have been directed at children in the twelve- to thirteen-year (seventh grade) age group or above, including the investigations reported by Evans et al., and Evans [13, 47]. The first of these reports describes some features of what Evans believed to be a promising social-psychological strategy to deter smoking in children. He reported that, from preliminary field work (interviews with a sample of 175 school children and teachers in grades five through seven), it appeared that three sources of pressure immediately influenced the child to begin smoking, these being pressure from peers, parents and media saturation pressure. His plan, therefore, was to first familiarize children with the nature of these pressures and then to specifically train them to cope with these. Within the context of such an approach, the disadvantages of smoking and the advantages of not smoking could be included but would not be the basic focus.

In their second paper, Evans et al. report on the implementation of these ideas. Their sample consisted of seventy-five seventh-grade students in the Houston Independent school district. The program used videotapes including information about the dangers of smoking, the advantages of non-smoking, the effects of smoking on other people, and a section describing and illustrating peer pressure, and its effects on smoking behavior. The influence of parents was also discussed and the video presented depiction of children's imitation of parents' smoking behavior, despite the admonitions of parents not to smoke. Finally, the pressures to smoke emanating from mass media were examined. At a ten-week follow-up, there were modest but significant differences between experimental and control groups. Based on reported smoking behavior, the treatment group and the control group, displayed smoking onset percentages of 10 and 18.3 percent respectively. These findings were essentially replicated in a larger study incorporating intervention and assessment over three junior high school years (Evans et al. cited in [48]).

Colquhoun and Cullen designed and implemented an anti-smoking program involving 297 Brusselton school children aged twelve to fourteen years [28]. The program, which was carried out in the classroom, consisted of six sessions and involved discussions on the general physiology of respiration and metabolism; the power of advertising and the profit motive of advertisers; smoking and health; the nature and value of friendships; the pressure on an individual to conform with peer groups, and general discussion. Results indicated that the program significantly reduced the number of children smoking at a one-year follow-up. Of special note is Colquhoun and Cullen's observation that the program's effects were greater for the youngest (twelve-year-old children) indicating how important it may be to commence anti-smoking efforts with children at or below this age level.

An entire class of ninth-graders (N = 149, aged approximately fourteen years) from an inner-city Chicago high school, were involved in a treatment condition featuring either role play and discussion, or only discussion, in a smoking prevention program designed and described by Jason, Mollica and Ferrone [49]. A further sixty-six same-age students from another Chicago high school served as controls. The subjects' role plays each emphasized a different negative aspect of smoking (immediate negative consequences perceived by onself, perceived by others, and long-term health hazards). Following role plays were discussions on topics related to smoking. In the discussion-only group, the focus was on any aspect of smoking (for example, reasons why people begin to smoke, health hazards associated with smoking, ways of reducing smoking). Four advanced psychology students led the groups and there were six weekly sessions.

The major finding of the Jason et al. study was that cigarette smoking could be significantly reduced among teenagers exposed to a role playing intervention [49]. The treatment was aimed at building competencies and skills to better enable children to avert the maladaptive behavior pattern. Specifically, results showed that of the initial smokers in the experimental group, over 50 percent were abstinent at posttest and at a three-month follow-up, while smoking had increased among control group subjects. A further important finding was that there were only minimal changes for regular smokers. Jason et al. concluded that it seems preventive projects are needed to catch incipient smokers before their habits have become well entrenched.

In an attempt to develop a broader and more comprehensive approach to smoking prevention, Botvin and Eng tested the efficacy of a ten-session school-based smoking prevention program designed to increase students' general personal competence, as well as their ability to deal with specific social influences to smoke [50]. Included were components dealing with: a) knowledge about the prevalence and immediate consequences of smoking, b) decision making, c) techniques for coping with anxiety, and d) basic social skills including pressure-resistance tactics. Their subjects were 281 eighth-, ninth-, and tenth-grade students in two New York schools, who met weekly with members of the project staff. Over a period of six months the psychosocial smoking prevention program tested in this study succeeded in reducing the incidence of new, experimental smoking by nearly 70 percent. Botvin and Eng concluded that, based on these results, it appeared that this type of smoking prevention strategy was effective in at least inhibiting or delaying the onset of cigarette smoking. However, their recommendation for future research was that a younger population should be utilized, since the critical time for the onset of cigarette smoking appeared to begin at an earlier age.

The purpose of a follow-up study by Botvin and Eng was to test the efficacy of this type of comprehensive multicomponent smoking prevention program when implemented by older peer leaders [51]. The underlying strategy behind the use of peer leaders was to harness the power of peer pressure in friendship and support groups, to reinforce non-smoking and other health-promoting behavior as "cool" and acceptable behavior [14]. More specifically, Botvin and Eng's study was designed to reinforce the effectiveness of this type of intervention and to determine [51]:

1. its effectiveness with a younger population;
2. its effectiveness when implemented by peer leaders;
3. the extent to which reductions in new smoking achieved at the end of the program could be sustained without any additional intervention activities; and
4. the extent to which reductions in new experimental smoking ultimately translate into reductions in regular cigarette smoking.

The subjects were 426 seventh graders from two schools in New York. The program ran for twelve sessions and there was a twelve-month follow-up.

Again the main emphasis in the Botvin and Eng program was on the learning of basic skills in order to improve general personal competence, as well as students' ability to cope with the various social pressures to smoke [51]. Material was taught through group discussions, modeling and behavioral rehearsal. The peer leaders conducting the program were older eleventh- and twelfth-grade students. Results showed that at posttest there were significantly fewer new smokers in the experimental compared to the control group. In addition to its effects on the onset of smoking, the experimental group had significantly higher posttest scores than the controls for smoking knowledge, psychosocial knowledge and advertising knowledge. At the one-year follow-up there was no significant difference in the numbers of new smokers in control and experimental groups. However, significantly fewer of the experimental group had smoked over the last week, i.e., the program resulted in a 56 percent reduction in new regular smoking.

Coe et al. also made use of older peer leaders to present non-smoking as an attractive, smart lifestyle, to the seventh-grade children in their study [14]. The study was conducted in two public middle schools in the Saint Louis, metropolitan area. One class (ranging from fifteen to twenty students) in each school was assigned to the experimental condition and one to the control. Eight sessions of one hour each were conducted by freshman students. The intervention strategies included sessions first on peer pressure to smoke. Opinions were shared in group discussions and the role-playing exercises were used to encourage students to develop the skills for effectively resisting such pressure. Secondly, students were encouraged to develop a skeptical view of cigarette advertising. Students analyzed cigarette ads to help them become aware of specific persuasion tactics used by tobacco companies, and discussed ways of resisting these advertising messages. Finally, an award was offered to the class most successful in reducing smoking as a positive reinforcement for not smoking. The results of this study revealed that the proportion of adolescents who remained non-smokers was higher among program participants than controls, both at the end of the program and one year later.

The paper by Telch et al. reported on the long-term follow-up data on the effectiveness of Project CLASP—a smoking prevention program for adolescents, employing high school students to guide younger peers in the development of counter-arguing skills to resist social pressures to smoke [4]. Subjects were seventh-grade students from two junior high schools near Stanford University. One school served as experimental, and the other served as control. Surveys were conducted three times yearly over the first two years in junior high school (seventh and eighth grade), and once during the third year (ninth grade).

Telch's high school peer-leader-teams, composed of five to seven school students, conducted all class sessions. Their training consisted of three two-hour sessions focusing on the use of basic behavior change skills. The program consisted of seven sessions over a nine-month period with the focus on public commitments not to become dependent on tobacco, learning some of the social influences which encourage young people to adopt the smoking habit and practicing ways to resist this pressure, becoming aware of promotional techniques which encourage smoking, and covering responses to counter these effects. Results showed that at a nine-month posttest there were significant differences between weekly smoking of controls (10.3 percent) and treatment groups (5.3 percent). At twenty-one months there was an even greater discrepancy, and at thirty-three months there continued to be a considerable significant difference. These findings lend support for the long-term effectiveness of the project CLASP. It seemed that subjects acquired the necessary skills with which to resist the social pressures to smoke.

A two-year educational intervention program was carried out by Vartianinen and colleagues to try to prevent the onset of smoking among thirteen- to fifteen-year-olds in the county of North Karelia in eastern Finland [52]. The program was a Finnish adaptation of the one developed by McAlister and Corokers at Harvard and Stanford Universities [53]. Peer leaders, students one or two years older than the subjects, were used to deliver the anti-smoking message. The program used role play and other active learning techniques to teach skills for resisting the social and psychological pressures to smoke, and to reinforce negative attitudes towards smoking. There was intensive intervention in two schools, involving ten hours of instruction. General, less concentrated intervention, of five hours' duration, was carried out in the other schools of North Karelia two of which were matched with, and compared to, the intensive schools. Two further matched schools from another county in eastern Finland served as controls. During the two years there were 897 children involved, 95 percent at the first follow-up and 88 percent at the second. Results showed that the proportion of smokers was significantly greater in the outside schools than in the intervention schools. Overall, the increase in smoking was smallest in the intensive intervention schools, and this was more than doubled in the control schools.

The purpose of the program reported by Arkin et al. was to prevent the onset of smoking [5]. Subjects were 3206 seventh graders from eight suburban U.S. Twin Cities. These researchers compared two approaches to smoking prevention. The first, a short-term influences curriculum, focused on the immediate consequences of smoking, both social and psychological, and on the immediate social influences which encourage smoking. The second, a long-term influences curriculum, focused on the long-term health risks of smoking, presenting factual information and encouraging students to actively participate in the learning process. The short-term programs were either peer led with media supplements, peer led without media supplements, or health educator led. Peer leaders were those in each class nominated because they were admired and respected. Long-term programs were all led by health educators. All programs involved five sessions covering a six-month period. Posttests were administered shortly after the last session and results revealed that, on average, the experimental programs reduced the smoking onset rate by 37.3 percent and the weekly smoking rate by 58.8 percent for baseline smokers. However, there were no significant differences between the short- and long-term influences program, nor between the adult led or peer led groups.

Interestingly, while there appeared to be no distinctions between the types of program when measures were taken shortly after the conclusion of these programs in the Arkin et al. study [5], Luepker, Johnson, Murray, and Pechacek, reported that these differences are apparent when measurements are taken over an extended time period [54]. These researchers reported on the long-term follow-up of a program initially described by Hurd et al. [21]. Their five-session program ran for eight months and initially involved 1526 seventh graders from the Minnesota Independent school district. Sessions were led by either college-age leaders or familiar and respected role models selected from among their peers. Students were taught first to resist the social pressures to smoke; secondly, to be aware of the immediate harmful effects of smoking; and thirdly, to model their behavior after that of non-smoking peer leaders and older role models.

Luepker et al.'s follow-up included 1081 of those students initially involved, three years later [54]. Their data indicated that, in the control school, seventh graders progressed inexorably to higher levels of cigarette usage. The education program taught exclusively by adults was initially effective. However, by the end of the second-year follow-up, smoking rates in that school were similar to those in the control school. In the school where peer leaders were involved in the teaching program, the effect at prevention was most striking. Lower smoking rates prevailed to the end of the ninth-grade year, although there was a weakening of the effect over time. These data suggested that a smoking prevention program based on short-term, primarily social influences related to smoking, and using same-aged peer leaders could deter cigarette smoking at least over a three-year period. This lower smoking level suggests that the education program was still active, though somewhat weakened, at the final testing. Luepker et al. concluded that they believed that a second program timed for ninth or tenth grade, may be necessary to sustain the long-term reductions in smoking that everyone is seeking [54].

While all of those studies reporting on the effects of smoking education programs reviewed so far have been directed at high school age children, a number have suggested that, in fact, education at an even earlier age would be beneficial. Many have discovered that regular smokers are not very receptive to intervention, and most have stressed the importance of deterring the initiation into smoking. Yet as we are aware, for many this initiation process occurs at a primary school level. In fact, it has been demonstrated that the health effects of smoking are already occurring in pre-adolescent children. Bewley, Halil, and Snaith have given a clearly documented report on the findings of their study investigating the prevalence of smoking by primary school children, which set out to determine whether a relationship between smoking and respiratory symptoms appeared at this early age [6]. The sample consisted of 7115 children aged from ten to eleven and one-half-years, in their final year of primary school in Derbyshire. Of these children, seven percent reported a morning cough, 23 percent reported a cough at some time of the day or night, and 4.5 percent reported the cough lasted for three months or more. These respiratory symptoms were all significantly related to the level of smoking.

Similarly, Bewley and Bland reported on a study of smoking and respiratory symptoms in 5355 final year primary, and first year secondary, school children in Kent, aged ten to eleven and one-half-years [7]. Morning cough was reported by 10 percent of the children, day or night cough by 26 percent and cough lasting three months by 7 percent. Once again, smoking was consistently associated with a higher prevalence of respiratory symptoms, and there was a dose-response relationship between the amount smoked and such symptoms. Since confirmed habitual smoking and early lung damage are known to cause severe respiratory disease in later life, Bewley and his colleagues believe the implications of their findings were therefore clear—efforts must be made to persuade children at primary school not to start smoking.

Unfortunately, so far the results of those programs directed at the younger age group are not too encouraging. The study reported by Pederson, Stennet, and Lefcoe constitutes the short-term post-program evaluation of a smoking education curriculum for children aged nine- to eleven-years [17]. Subjects were ninety-nine grade-four children and 101 grade-six children from eight public schools in Ontario. Half of the classrooms for each grade were exposed to the smoking awareness program by the teacher. Twelve classroom hours were spent on the material in the Chicago Lung Associations' publication "The Respiratory System and Smoking—A Complete Teaching Curriculum," and on other topics related to cigarette smoking, i.e., smoking and fitness, why people start smoking, decision-making and smoking, factors to consider about smoking (cost, smell, stain, fire danger, fatigue), and difficulties in quitting smoking. Results showed an improvement in children's attitudes towards, and knowledge of, the effects of smoking. Pederson, Stennet, and Lefcoe claimed that the failure to find any impact of the program on children's smoking behavior was expected, since very few (less than 5% of the total sample) were regular smokers. They concluded that it was possible that their program could prevent the children from becoming regular smokers over the following few critical years, but at that point an estimate of its effectiveness could not be made.

The long-term effectiveness of this program was, however, tested soon after, when half of the children from twenty of the sixty-three public schools in London, Canada, were exposed to the same curriculum guide reported above [24]. The other half served as controls. Initial testing was conducted on 2583 children in grades four to six in 1975. The second testing involved 2372 children in grades six to eight. Results were available for 1682 children who responded on both occasions, and showed that slightly more than half of the students reported no change in smoking habits over the three years. A large majority of those reporting a difference, increased their involvement with cigarettes. This trend was not altered by exposure to anti-smoking education.

Pederson, Baskerville, and Lefcoe believed that the essential ingredients missing from their program were strategies for dealing with peer pressure, which they felt must be included in curricula designed to combat smoking [24]. Also worth special mention is their observation that among children studied, those who had experimented with cigarettes in 1975, were much more likely than the non-smokers, to be smokers in 1978. Pederson et al. concluded that because of the importance of earlier smoking status in determining later status, education efforts may have an effect if begun very early in the child's schooling. They believed that attempts to prevent any experimenting with cigarettes before grade four could reduce later regular smoking, and that programs aimed at adolescents and teens are probably too late to have any significant effect on those children who are fairly likely to become smokers.

A "Smoking or Health" program was developed by the New South Wales Department of Education in an attempt to modify primary school children's knowledge, attitudes, intentions, and behavior concerning smoking [55]. The project involved 6000 children aged between ten and twelve years. The actual program contained nine different strategies ranging from presentation of some elementary physiological facts about smoking and its effects on health activities which helped children clarify their values with respect to health-associated behavior; materials aiming to demystify cigarette advertising; other activities which helped children develop skills in countering peer group pressures while preserving their self esteem; and the use of role play to simulate behaviors concerned with decision making about whether to use or not use cigarettes. The class teacher implemented the program after having attended a one-day inservice training course conducted by a local health education consultant. One and a half hours for each of six weeks were allotted, with three one and a half-hour review sessions following that.

The results of the Lloyd et al. study failed to demonstrate any substantial effect from the program. Smoking became more prevalent in all subgroups of children during the study period. There were significant increases in the children's knowledge of the health hazards associated with smoking. However, this increased knowledge did not result in behavior change in the desired direction. Lloyd et al. did, however, make special mention of the fact that the results they reported were of the short-term outcome only, and noted that longer-term evaluation may be necessary in order to give a complete picture of the effectiveness of the program [55].

According to Schinke and Gilchrist, past work paves the way for prospective school prevention [56]. Findings that tobacco habits are launched in junior high, call for prevention programs aimed below grade seven. In their study, fifty-six sixth graders from two Seattle schools were the participants (mean age 11.7 years). Their treatment included eight one-hour semi-weekly sessions, delivered by two peer group leaders. Group members watched and discussed films giving the pros and cons of smoking. Students acquired a systematic approach to problem solving and decision making, and practiced sticking to decisions in role plays first modeled by leaders. This continued until students were able to withstand pro-smoking pressures, urges and temptations. Subjects were posttested after two months, with a follow-up six months later.

The results of the Schinke and Gilchrist study were more promising than the Pederson et al. [17, 24] and Lloyd et al. [55] studies which had utilized teachers as the health educators. Knowledge test scores were higher for students given primary prevention than for those left untrained. They were also able to take more perspectives on tobacco use problems and to link problems with solutions. In addition, they had superior insights when predicting others' reactions to non-smoking decisions, and evidenced greater self control during videotaped interactions. On follow-up questionnaires, primary prevention students, in contrast to control students, expressed stronger commitments to abstaining from tobacco, and noted more instances of refusing all forms of tobacco. Trained youths also reported less tobacco use. Halfway through seventh grade, 8 percent of primary prevention, and 37.5 percent of control students had smoked in the previous month. Schinke and Gilchrist concluded that primary prevention is a cost-effective method to delay the onset of smoking, but that if health educators hope to reduce tobacco consumption, they must reach children before smoking is habituated.

In summary, it appears that prevention remains our principal weapon in the fight against tobacco-induced disease. Because of this, numerous smoking prevention programs aimed at school age children have been designed and implemented. Up until the late 1970s, almost all of these programs had focused on providing children with factual information on the hazards of cigarette use, and low-key non-directive approaches had been utilized. In general, such programs showed little success. More recently, interventions have concentrated more on equipping children to resist social pressures towards cigarette use, and these programs have been much more successful.

Almost all anti-smoking programs implemented within the last five years have been directed at children in the twelve- to thirteen-years age group or above. Most of these have been quite successful at reducing smoking levels among the students, especially those utilizing peers as group leaders. However, many of the researchers evaluating such programs have suggested that it may be valuable to use younger children as targets, since the main goal is to prevent smoking onset. For many, this occurs at the primary school level, where its physical effects are already becoming apparent. Also, the more regular smokers found among high school students have been found to yield less to smoking behavior changes.

To date, there have been few reports of smoking prevention programs directed at children in the below twelve-years age group. Of the three cited here, two, utilizing teachers as program leaders, were unsuccessful at changing the children's smoking behavior. Results of the third, which made use of peer leaders, were more encouraging. It appears, therefore, that while it seems important to begin smoking education during primary school years, more attention needs to be given to the observation that anti-smoking messages are perhaps best delivered by those who play a more major role in the smoking initiation process.

PARENTS AS EDUCATORS

While most parents want their children to receive the best education possible, too many of them leave this entirely to the schools, little realizing that they can play a vital role in it [57]. During the last decade there has been an increased awareness of the importance of including parents in the educational process, especially where younger children are concerned [58], for as O'Dell has pointed out, it is the parents who are the primary source of influence during the child's formative years [59]. Over the last few years educational programs have attempted to find ways to teach parents new skills, to provide them with counseling, to have them become teachers of their own children, to use them as volunteers in the classroom, and to participate in many other types of parent involvement [60].

Parent volunteers can be trained to be very effective aides for teachers according to Benson and Russ [61]. This author reports on a project which took place in a primary school classroom for thirteen trainable mentally retarded children in Cicero, Illinois. The children were aged from five to eight years. The program's first goal was to provide the children with more individualized instruction, the second was to promote parental interest and involvement in their children's educational growth, and the third was to encourage and develop community interest and support. The program was considered to be very successful and much improvement was noted in areas such as self-care skills, academic skills, and fine-motor skills.

Numerous home tutoring programs have also been reported and it is common knowledge that parents frequently attempt to assist their children with homework and other academic tasks. The study carried out by Broden, Beasley, and Hall sought to assess the effects of a spelling tutoring program carried out at home by a parent on weekly spelling tests [62]. Results showed that in-school spelling scores increased significantly. Similarly, Petrie, Kratochwill, Bergan, and Nicholson evaluated the effects of a training program designed to teach parents skills for facilitating the academic performance of their children [63]. Their results revealed a substantial increase in the targeted children's printing performance. Such findings have led Gordon to conclude that the parent is the primary teacher of the child, and can enhance the child's learning and in fact his whole development, by providing a learning environment, by modeling behavior, and by engaging in direct instruction with the child [64].

Although few health education programs have made use of the potential force of a child-parent interaction to accomplish the program goals, there is a growing recognition that parents can be the "potent force" in preventing drug abuse. This belief has been given a great deal of support [65].

The importance of the peer group and family in relation to smoking emphasizes the need for programs to aim at the wider social milieu of the child. As previously mentioned, recent approaches have attempted to address the recognized importance of peer pressure by teaching children how to resist this pressure, and, in a number of cases, by using peer leaders as role models. However, starting to smoke is a multi-step process, with different influences operating at different ages according to Wake (cited by Purcell et al. [38]). In the years of middle childhood (six to twelve years), curiosity and parental modeling may exert the most influence on whether a child smokes or not. During adolescence, peer group pressures and the beginnings of independence from their parents begin to exert a greater influence on behavior. Therefore, if smoking prevention is to be aimed towards pre-pubescent children, as previously recommended, then educational programs should involve the parent population which could be utilized as the source of smoking education.

Up until now, parents have largely remained an untapped resource with respect to children's smoking education. In fact, only one-third of the 787 nine-year-old Dunedin school children sampled in 1982 had even discussed cigarette related matters with their parents [11]. When parents had initiated such discussions, it was often simply to tell children not to smoke, or to inform them of the harmful health effects of smoking. Based on these findings Oei et al. recommended that parents should be encouraged to utilize their significant influence to play a more major role in directing their children towards a non-smoking stance. Similarly, Hardes and colleagues also felt that parental involvement in smoking education should be increased, when such education was to be directed at children in the younger age group [66]. From their evaluation of the "Smoking or Health" program administered to the ten-year-old New South Wales school children they concluded that there was a need for preventive measures at an earlier age than that they had attempted. They believed that this should be supplemented by increasing parental awareness of early introduction to, and use of, the more commonly accepted drugs.

The limits of restricting children's smoking education to the school setting have been recognized by Sunseri et al. [67]. What they felt was needed was a comprehensive program that involved both family members and students that provided factual information about the deleterious effects of smoking on health; and that addressed both the social and psychological factors influencing adolescents' attitudes and their initiation of smoking behavior. Non-smoking modules incorporating these components were taught by teachers in three forty-five-minute sessions. In addition family non-smoking materials were sent home each week and family participation was encouraged. The subjects were 1143 public school students aged eleven to twelve years, and were assigned to either control group, program with family participation group, or program without family participation group.

As already discussed, parental influence is perhaps best harnessed at the nine-year-old level or below, when its force is most potent. Nevertheless, even among these eleven- to twelve-year-olds, the advantages of involving parents were apparent. Sunseri et al.'s results indicated that attitudes towards, and knowledge of the effects of cigarette use, were significantly improved only among those in groups with parental input [67]. While this teacher-led program had little effect on the actual smoking status of the children, those in the parental participation group were much less likely to purchase cigarettes than their counterparts.

Gordon and Haynes have also commented on the value of family involvement in smoking education [68]. However, their study involved the use of fourth-grade (approximately nine-year-old) children as a vehicle for influencing the smoking behavior and attitudes of a second target group, their parents, as opposed to vice versa. Of particular interest to them was whether a fourth-grade smoker education program could be used to induce cigarette smoking parents to reduce their smoking. Their study implemented the recommendation made in the 1979 Surgeon General's Report, that "further research should be conducted into school-based programs designed to influence parental lifestyles, including an assessment of the influence of such programs on smoking behavior."

Two samples of fourth-grade students participated in the study. The first, called the school sample, had the regular state mandated smoker education unit in which there were no homework assignments ($N = 273$). The second, the school-home sample, had the smoker education unit redesigned to include homework assignments including parents ($N = 502$). Gordon and Haynes commented that the fourth grade had been selected for study, since beyond this level children increasingly begin to experiment with cigarettes [68]. Questionnaires were sent out to parents six and a half months after program completion, in order to assess their attitudes towards smoking, and actual smoking behavior.

Comments from the parents regarding the anti-smoking homework were predominantly favorable and in many instances enthusiastic. Results showed that cigarette smoking parents who participated in the school-home smoker education program, in contrast to those who did not, reported smoking fewer cigarettes at follow-up, and also reported a greater decrease over that period of time in the amount smoked. Parents in the school-home sample were more adverse to their own children starting to smoke at a later time, but also tended to have more confidence in their children's judgment in this matter. Parent participants were also more inclined to favor the introduction of smoker-education as early as fourth grade, although the large majority of parents in both samples endorsed this practice.

Based on their findings, Gordon and Haynes concluded that the use of homework assignments as a technique for influencing the use of, and attitudes towards cigarettes among parents of fourth-grade children is a novel and practical approach to smoker education. It is easily able to reach a large target population and is likely to be reasonably well received by parents. It is easily introduced into existing health education programs and is economical. However, they did add that while the focus of interest in their study was on the smoking parent, the future cigarette smoking of the elementary school child is a matter of equal, if not greater, concern to health educators. The influence of such a program on children of this age group is yet to be assessed. The efficacy of a multi-component, parent-implemented smoking education program in reducing the rate of onset among primary school children, and in strengthening anti-smoking attitudes was recently investigated by us. The children involved were 362 nine-year-olds, chosen from fifteen Dunedin primary schools. The experimental group consisted of eighty-three children who completed the five-week smoking education program implemented by their parents. The remaining 171 children formed comparison groups. The general trend of results suggested that smoking onset rate and yearly and monthly smoking rates had all decreased following exposure to the program, but this trend was not significant. Attitudes towards smoking did become significantly more negative as a result of the program among those classified as puffers. Among non-puffers in the group exposed to the program, attitudes remained stable at the anti-smoking end of the scale. Parents involved in the education process had been given the opportunity to comment on it, and were generally impressed with the program content and were pleased to have been given the opportunity to be involved.

The main aim of the parent-led program implemented was to prevent the onset of smoking among children in the study. It was, therefore, considered important to measure naturally occurring changes in smoking status over time, as compared to changes among the experimental group. However, from the results, it appeared that smoking onset for this age group occurred at a very slow rate, with few children becoming new smokers in any of the comparison groups over the five-month period of data collection. A perennial question which confronts health educators dealing primarily with youth is: What kinds of outcomes are most meaningful to pursue in design and evaluation, if the target population displays a relatively low initial incidence of the health behavior which the education program is intended to shape [25]. In the current study, the percentage of puffers did increase slightly for control and refuser groups, compared with part- and full-program groups, until the follow-up. However, the small number of new smokers overall, meant this trend was not statistically significant. It seems that, in order to determine whether the educational messages had operated as a preventive measure, a more longitudinal study is needed.

Despite the fact that the differences between smoking onset rate among experimental and comparison groups were not marked, they were at least in the direction expected as a result of a successful education program. This was also true on other measures taken. While differences were not significant, there were substantial decreases observed in past-year and past-month puffing among program children. One possible explanation for this trend to have also been apparent among comparison groups, is the peer-influence issue. Since all participants in the study were classmates, it seems likely that new knowledge and changed attitudes and behavior among experimental group children, may have made a favorable impression on others in their peer group. A further possibility is that invitations to parents to become smoking educators, prompted discussions related to smoking within the homes of those in the part-program and refuser groups. Perhaps these had pay-offs in terms of children's reduced cigarette use. As with changes in smoking status, the assessment of the program's success in changing more regular use of cigarettes, obviously needs to span a longer time period.

Anti-smoking attitudes and beliefs are important factors in the onset of smoking, as has already been noted. Because of this, and since such attitudes are established at an early age and generally remain unchanged during secondary schooling [29], another important aim of the Smoking Education Program, was to encourage the formation of more definite anti-smoking attitudes among these primary schoolers. Results indicated that the program's success at achieving this goal varied as a function of previous puffing experience. It seemed that the attitudes of the non-puffers were already within the desired direction prior to intervention, and thus the program could do little more than reinforce the beliefs of these children. Among puffers, however, attitudes towards smoking did become significantly more negative following exposure to the program. This finding was especially important since, in comparison, members of the control group had become more accepting of the habit over the five-month period. In this respect, at least, the goals of the intervention had been achieved.

While the parental responses collected from those involved in the smoking program were of interest, it would be misleading to draw any conclusions from the findings since the sample was biased with the observation that mothers more frequently took on the role of health educator, irrespective of smoking status. Also worthy of mention was the reduction in mother's daily consumption of cigarettes among the program group. This finding was consistent with Gordon and Haynes's report of reduced cigarette intake among those parents involved in helping their children with smoking education homework assignments [68].

That parents are valuable resources and crucial agents in the health education process is now becoming more widely accepted. But how willing are parents to take on this responsibility? How accessible are they? The present study has demonstrated that, in fact, parents are a cost-efficient, time-efficient, very effective resource, who are easily accessible. When asked to comment on having been given the opportunity to act as educators, and to give opinions on the program's content, the large majority of parents were impressed with the idea and stated that they felt a lot had been gained from becoming involved. At the very least, such involvement promoted open discussions regarding smoking, which otherwise do not seem to occur between parents and children in this age group [16].

Reviewing the factors which contribute toward children's decisions of whether or not to try smoking, provides good indication of the direction which future smoking education programs need to take. As the aim of smoking education programs is to prevent smoking onset, it is now widely accepted that such interventions should begin during the primary school years. The influence that the smoking habits of friends and family have on a child's decision to smoke highlights the need for education programs to consider not only the individual, but also his or her wider social environments.Children need to be made aware of the effectiveness of models; parental, peer, and media, in influencing their decision to try smoking. They additionally need to learn to understand and deal with such factors. Knowledge of both long- and short-term effects of smoking, and the immediate advantages of being a non-smoker also seem to be considered important, suggesting other components in an effective education package.

Starting to smoke is a multi-step process with different influences exerting more power at different stages of development, and a spiral program, covering a number of years to ensure continuity and reinforcement would seem to be useful. Smoking prevention programs involving students around the age of twelve years using peer leaders as health educators have been reasonably successful at reducing smoking rate and onset, though it was apparent that education at an even earlier age would be more effective. During the primary school years it is the parents who are the primary source of influence. Despite this, few health education programs have made use of the parent-child interaction. This review demonstrated the potential of parents as a potent force in a smoking education prevention program. Future researchers must now pay more attention to the call by Dupont, the former Director of National Institute of Drug Abuse, who said that "The most important new frontier in primary prevention of drug abuse is parent power. It is ironic that after a decade of parent put-downs, we are rediscovering that parents, who were written off as ignorant and meddlesome at best, and as the problem at worst, are now the solution to the drug problem" (cited in Cohen [65]).

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